

4. RETAIL AND SALES

Domain: Retail and Sales

Project Title: Store Sales Trend Tracker with Visual Reports

Problem Statement: Record daily sales transactions with product details, quantity, and revenue. Use Pandas to calculate sales totals, identify peak days, and generate pivot table summaries. Visualize category-wise sales using pie charts and top products using bar charts

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CODING:

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
class SalesAnalyzer:
```

```
    def __init__(self, file_path):
```

```
        self.df = pd.read_csv('/content/retail_sales_dataset.csv')
```

```
        self.df['Date'] = pd.to_datetime(self.df['Date'])
```

```
        self.df['Month'] = self.df['Date'].dt.month
```

```
        self.df['DayOfWeek'] = self.df['Date'].dt.day_name()
```

```
        self.df['Quarter'] = self.df['Date'].dt.quarter
```

```
    def get_sales_summary(self): # sales summary
```

```
        total_revenue = self.df['Total Amount'].sum()
```

```
        avg_transaction = self.df['Total Amount'].mean()
```

```
        total_transactions = len(self.df)
```

```
        total_quantity = self.df['Quantity'].sum()
```

```
        summary = {
```

```
            'Total Revenue': total_revenue,
```

```
            'Average Transaction Value': avg_transaction,
```

```
            'Total Transactions': total_transactions,
```

```
            'Total Quantity Sold': total_quantity
```

```
        }
```

```
        return summary
```

```
    def get_daily_sales(self): # daily sales
```

```
        daily_sales = self.df.groupby('Date')['Total Amount'].sum().reset_index()
```

```
        return daily_sales
```

```
    def get_peak_days(self, n=10): # peak sales
```

```
        daily_sales = self.get_daily_sales()
```

```
peak_days = daily_sales.nlargest(n, 'Total Amount')
return peak_days
```

```
def create_category_pivot(self):
    pivot_table = pd.pivot_table(
        self.df,
        values='Total Amount',
        index='Product Category',
        columns='Month',
        aggfunc='sum',
        fill_value=0
    )
    return pivot_table
```

```
def create_gender_pivot(self):
    gender_pivot = pd.pivot_table(
        self.df,
        values='Total Amount',
        index='Product Category',
        columns='Gender',
        aggfunc='sum',
        fill_value=0
    )
    return gender_pivot
```

```
def plot_category_sales(self):
    category_sales = self.df.groupby('Product Category')['Total Amount'].sum()
    plt.figure(figsize=(5, 5))
    plt.pie(category_sales, labels=category_sales.index, autopct='%1.1f%%')
    plt.title('Sales Distribution by Product Category')
    plt.show()

    return category_sales
```

```
def plot_top_products(self, n=10):
```

```
top_categories = self.df.groupby('Product Category')['Total Amount'].sum().nlargest(n)
```

```
plt.figure(figsize=(6, 6))
top_categories.plot(kind='bar')
plt.title(f'Top {n} Product Categories by Revenue')
plt.xlabel('Product Category')
plt.ylabel('Total Revenue')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
return top_categories
```

```
def plot_daily_sales_trend(self):
```

```
    daily_sales = self.get_daily_sales()
```

```
    plt.figure(figsize=(9, 7))
    plt.plot(daily_sales['Date'], daily_sales['Total Amount'])
    plt.title('Daily Sales Trend')
    plt.xlabel('Date')
    plt.ylabel('Total Sales')
    plt.grid(True)
    plt.tight_layout()
    plt.show()
```

```
    return daily_sales
```

```
def plot_monthly_sales(self):
```

```
    monthly_sales = self.df.groupby('Month')['Total Amount'].sum()
```

```
    plt.figure(figsize=(6, 6))
    monthly_sales.plot(kind='bar')
    plt.title('Monthly Sales Summary')
    plt.xlabel('Month')
    plt.ylabel('Total Sales')
```

```
plt.xticks(rotation=0)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
return monthly_sales
```

```
def plot_gender_sales(self):
```

```
    gender_sales = self.df.groupby('Gender')['Total Amount'].sum()
```

```
    plt.figure(figsize=(6, 6))
```

```
    plt.pie(gender_sales, labels=gender_sales.index, autopct='%1.1f%%')
```

```
    plt.title('Sales Distribution by Gender')
```

```
    plt.show()
```

```
    return gender_sales
```

```
def plot_age_group_sales(self):
```

```
    self.df['Age'] = pd.to_numeric(self.df['Age'], errors='coerce')
```

```
    bins = [0, 20, 30, 40, 50, 60, 100]
```

```
    labels = ['<20', '20-30', '30-40', '40-50', '50-60', '60+']
```

```
    self.df['Age Group'] = pd.cut(self.df['Age'], bins=bins, labels=labels)
```

```
    age_sales = self.df.groupby('Age Group')['Total Amount'].sum()
```

```
    plt.figure(figsize=(6, 6))
```

```
    age_sales.plot(kind='bar')
```

```
    plt.title('Sales by Age Group')
```

```
    plt.xlabel('Age Group')
```

```
    plt.ylabel('Total Sales')
```

```
    plt.tight_layout()
```

```
    plt.show()
```

```
    return age_sales
```

```
def plot_scatter_sales(self):
```

```
plt.figure(figsize=(8,6))
plt.scatter(self.df['Quantity'], self.df['Total Amount'], alpha=0.6)
plt.title("Scatter Plot: Quantity vs Total Amount")
plt.xlabel("Quantity")
plt.ylabel("Total Amount")
plt.grid(True)
plt.show()
```

```
def plot_histogram_sales(self):
    plt.figure(figsize=(8,6))
    plt.hist(self.df['Total Amount'], bins=30, alpha=0.7, edgecolor='black')
    plt.title("Histogram of Total Sales Amount")
    plt.xlabel("Total Amount")
    plt.ylabel("Frequency")
    plt.show()
```

```
def plot_kde_sales(self):
    plt.figure(figsize=(8,6))
    sns.kdeplot(self.df['Total Amount'], fill=True)
    plt.title("KDE Plot of Total Sales Amount")
    plt.xlabel("Total Amount")
    plt.ylabel("Density")
    plt.show()
```

```
analyzer = SalesAnalyzer('/content/retail_sales_dataset.csv')
```

```
summary = analyzer.get_sales_summary()
```

```
print("Sales Summary:")
```

```
for key, value in summary.items():
```

```
    print(f'{key}: {value:,.2f}')
```

```
# Get peak days
```

```
peak_days = analyzer.get_peak_days(5)
```

```
print("\nTop 5 Peak Sales Days:")
```

```
print(peak_days)
```

Create pivot tables

```
category_pivot = analyzer.create_category_pivot()  
print("\nCategory-wise Monthly Sales Pivot Table:")  
print(category_pivot)
```

```
gender_pivot = analyzer.create_gender_pivot()  
print("\nCategory-wise Gender Sales Pivot Table:")  
print(gender_pivot)
```

Category sales pie chart

```
category_sales = analyzer.plot_category_sales()
```

Top products bar chart

```
top_products = analyzer.plot_top_products()
```

Daily sales trend

```
daily_trend = analyzer.plot_daily_sales_trend()
```

Monthly sales

```
monthly_sales = analyzer.plot_monthly_sales()
```

Gender sales distribution

```
gender_sales = analyzer.plot_gender_sales()
```

Age group sales

```
age_sales = analyzer.plot_age_group_sales()
```

Scatter plot

```
analyzer.plot_scatter_sales()
```

Histogram

```
analyzer.plot_histogram_sales()
```

KDE plot

analyzer.plot_kde_sales()

OUTPUT:

Sales Summary:

Total Revenue: 456,000.00

Average Transaction Value: 456.00

Total Transactions: 1,000.00

Total Quantity Sold: 2,514.00

Top 5 Peak Sales Days:

Date	Total Amount
134 2023-05-23	8455
127 2023-05-16	7260
166 2023-06-24	6220
44 2023-02-17	5890
203 2023-08-05	5205

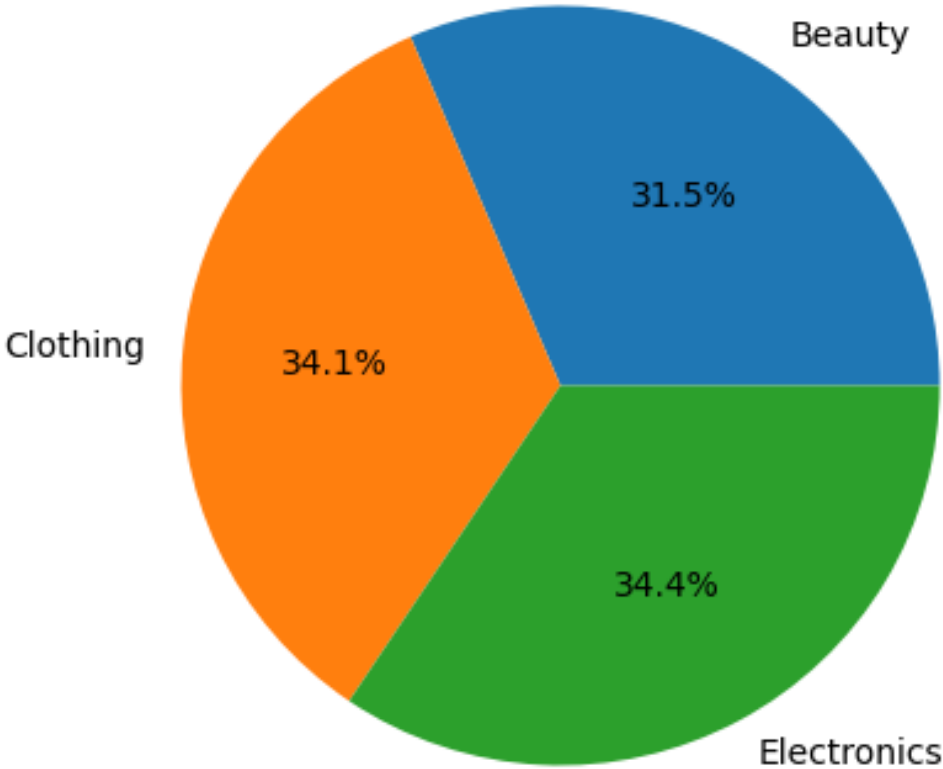
Category-wise Monthly Sales Pivot Table:

Month	1	2	3	4	5	6	7	8	\
Product Category									
Beauty	13930	14035	10545	11905	12450	10995	16090	9790	
Clothing	13125	14560	15065	13940	17455	10170	8250	12455	
Electronics	9925	15465	3380	8025	23245	15550	11125	14715	
Month	9	10	11	12					
Product Category									
Beauty	6320	15355	9700	12400					
Clothing	9975	13315	15200	12070					
Electronics	7325	17910	10020	20220					

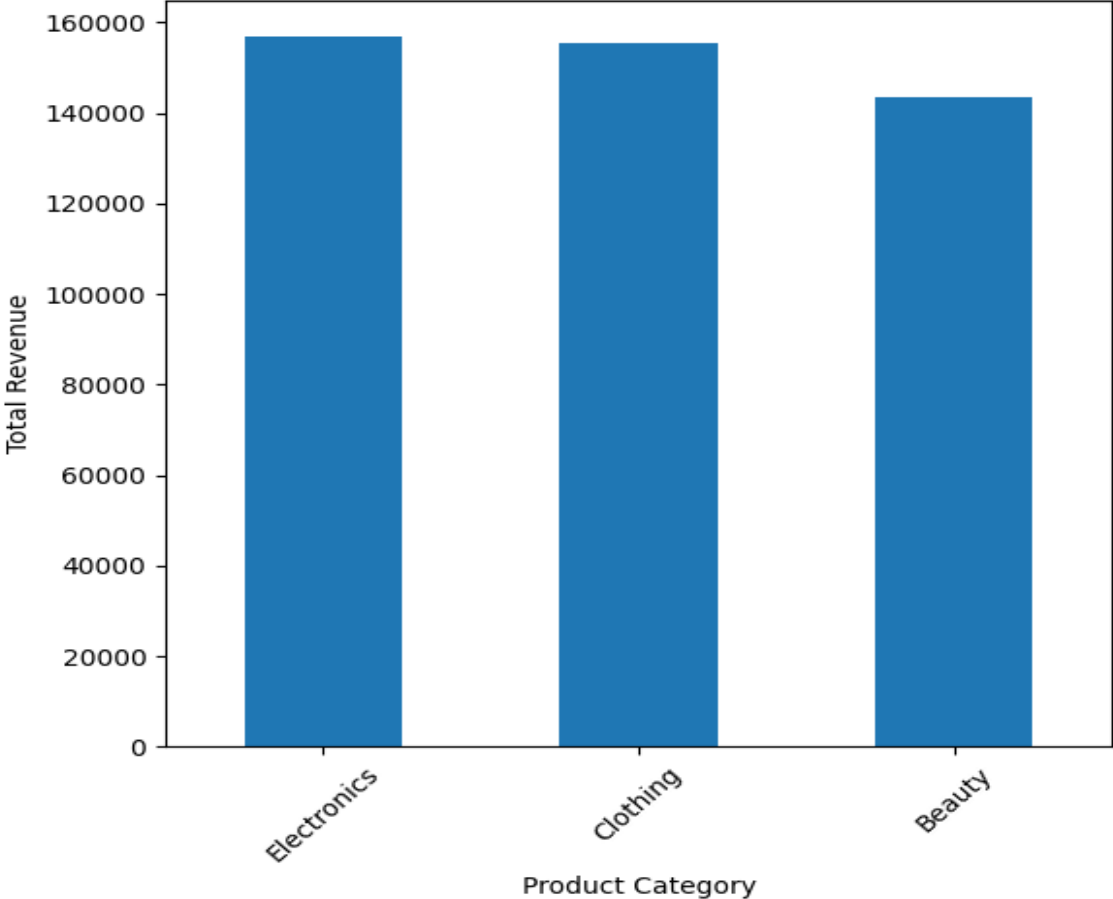
Category-wise Gender Sales Pivot Table:

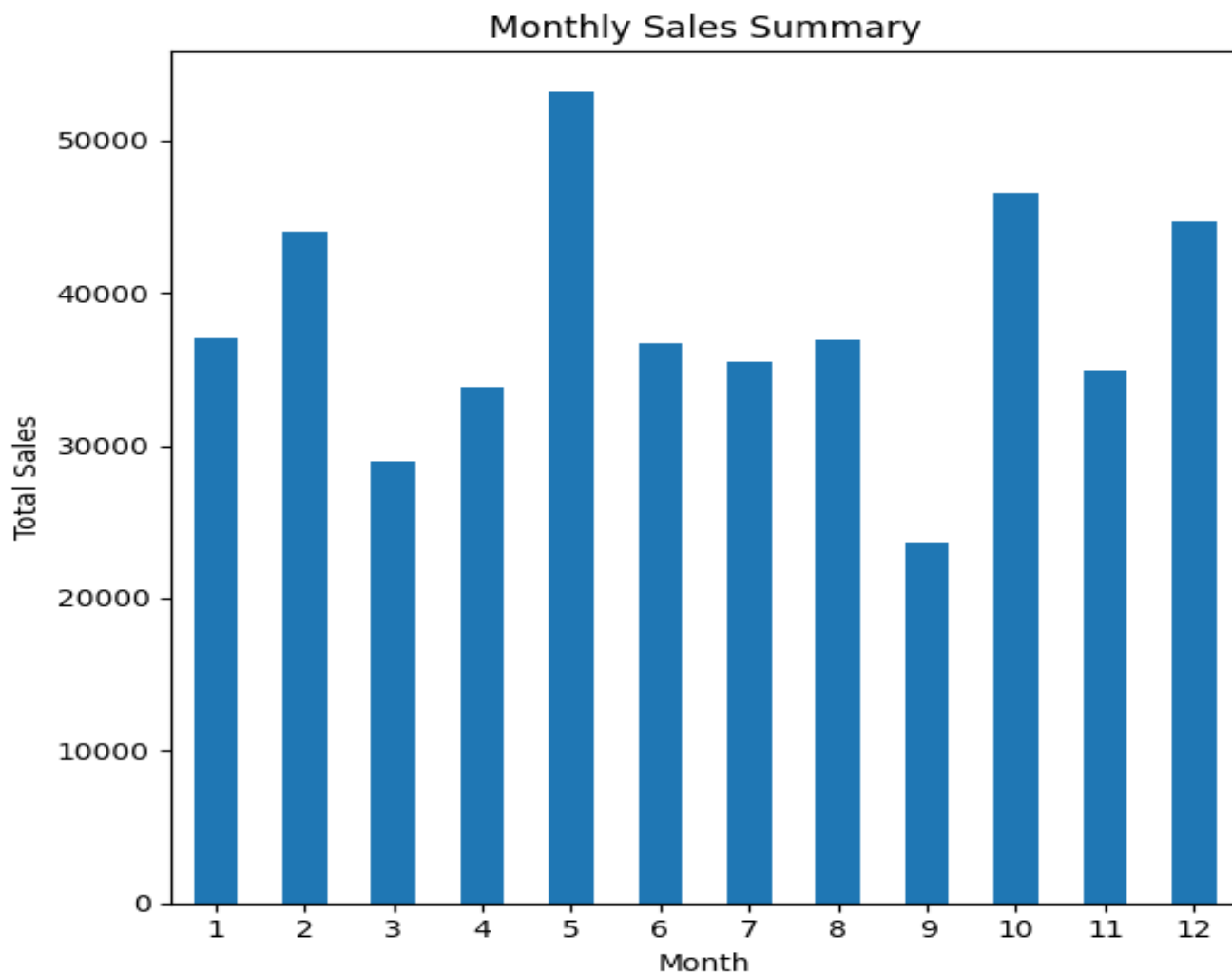
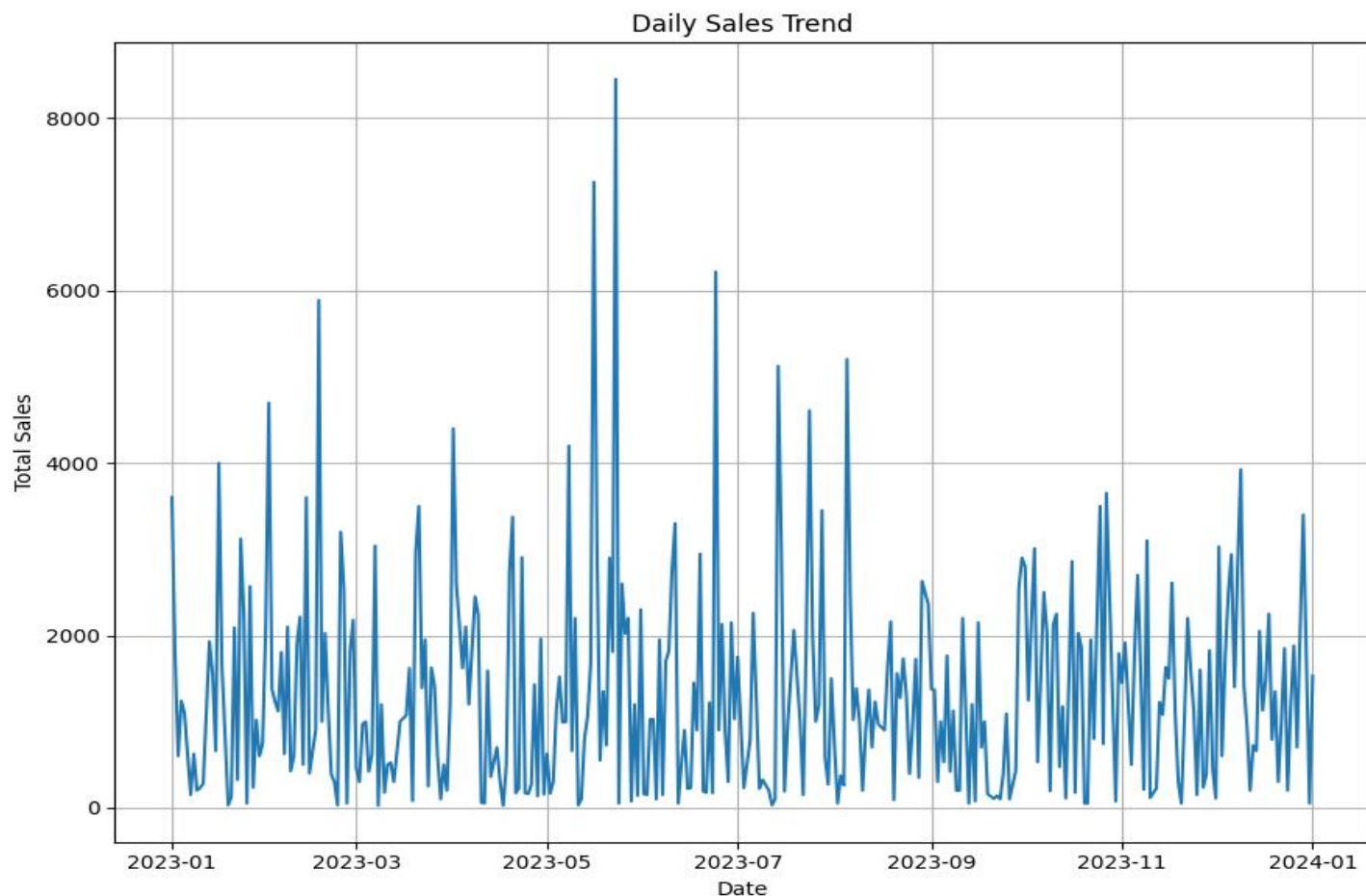
Gender	Female	Male
Product Category		
Beauty	74830	68685
Clothing	81275	74305
Electronics	76735	80170

Sales Distribution by Product Category

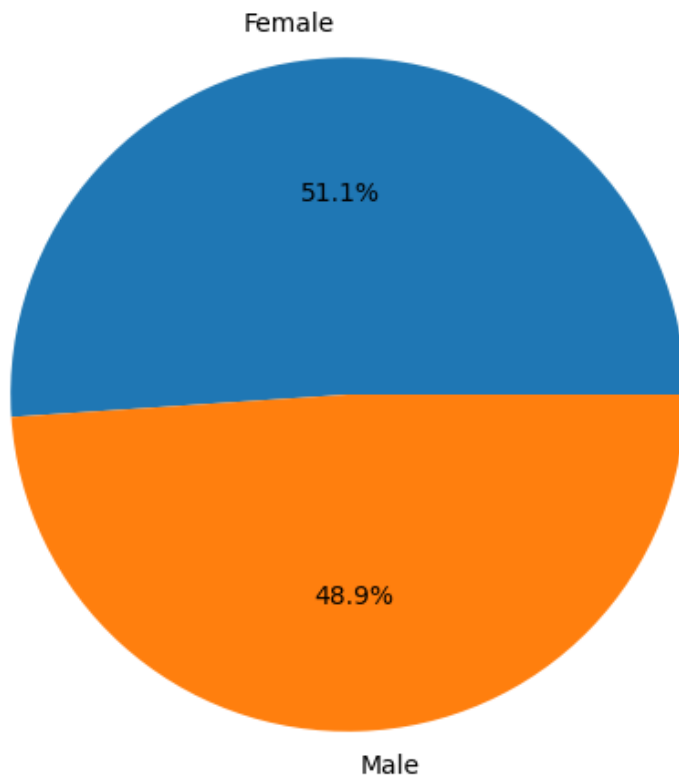


Top 10 Product Categories by Revenue

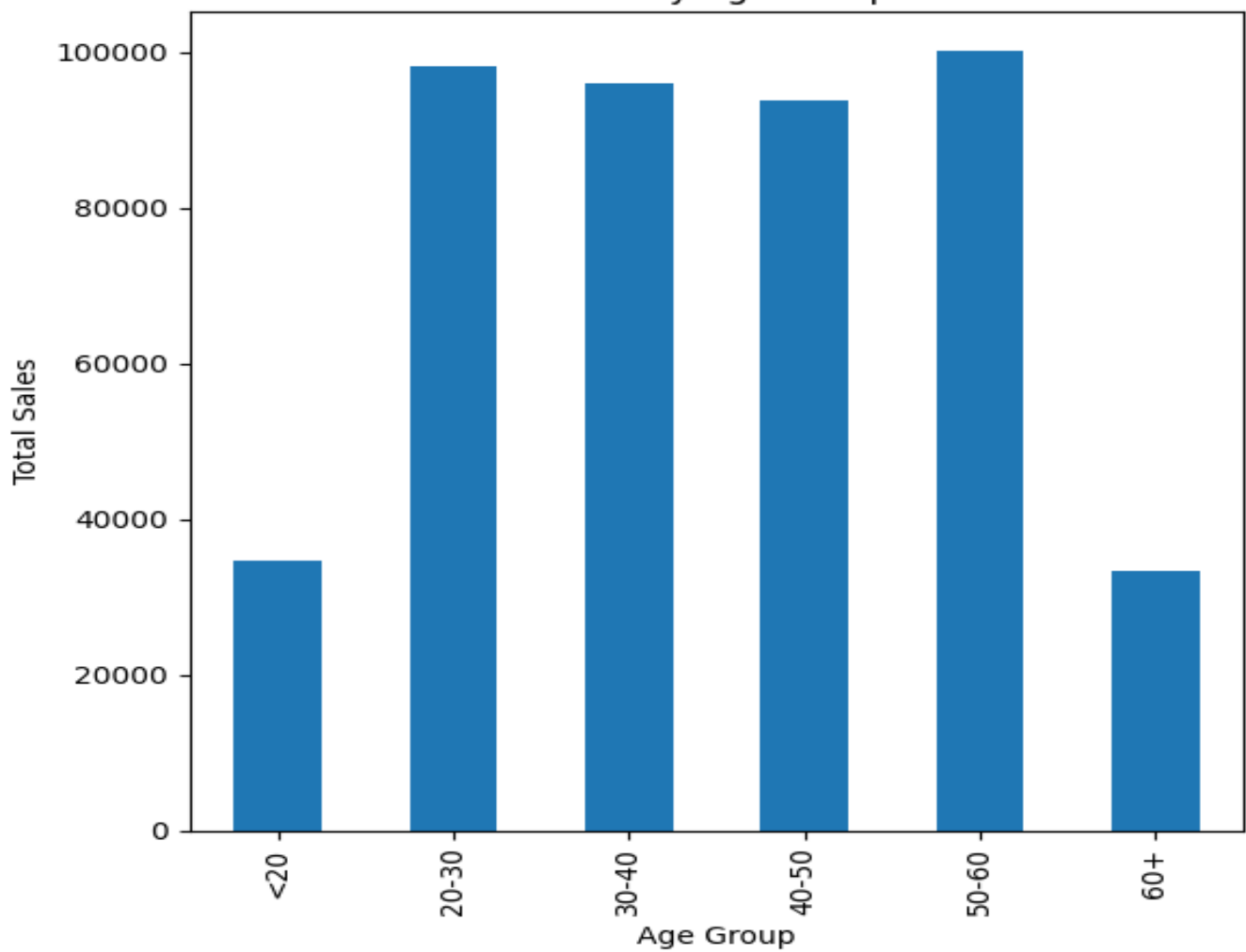




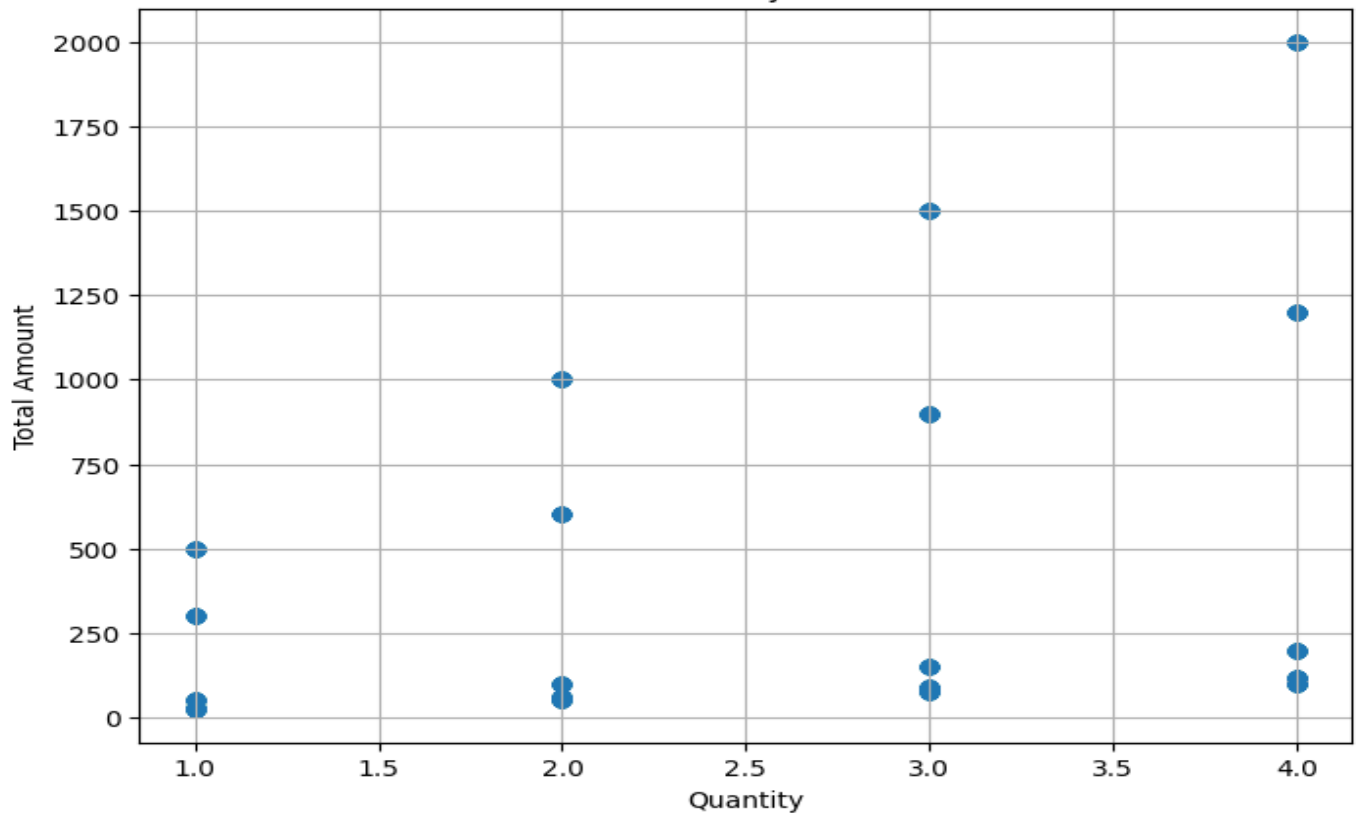
Sales Distribution by Gender



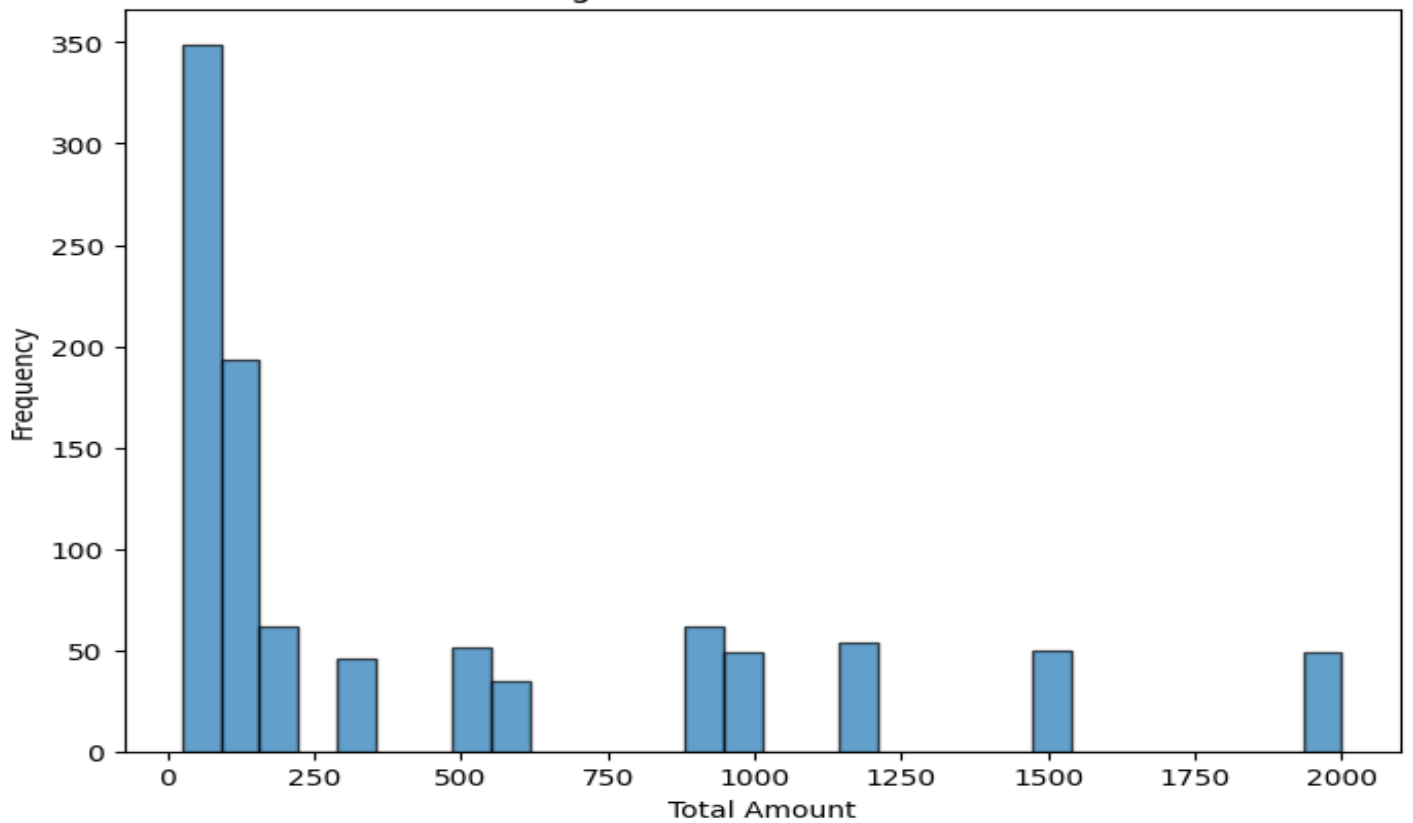
Sales by Age Group



Scatter Plot: Quantity vs Total Amount



Histogram of Total Sales Amount



KDE Plot of Total Sales Amount

